



DEPARTMENT OF INFORMATION SCIENCE & ENGINEERING

MAJOR PROJECT REVIEW (22IS76)

Temporal Spiking Graph Convolutional Networks for Low Power Dynamic Transaction Fraud Detection

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Abstract

Dynamic financial and network graphs require efficient methods for detecting fraudulent and anomalous activity. TS-GCN combines graph message passing with Leaky Integrate-and-Fire (LIF) spiking neurons to introduce temporal and event-driven processing. The model will be evaluated against a conventional GCN through classification performance, spike sparsity, latency, and estimated energy efficiency, with potential deployment on neuromorphic hardware.



Introduction

- Dynamic transaction and network data can be represented as graphs of entities and their interactions.
- GNNs learn these relationships but require continuous computation.
- SNNs use spikes and temporal neuron states for sparse, time-dependent processing.
- TS-GCN combines GNN-based spatial learning with SNN-based temporal processing.
- The model will be tested through software simulation and evaluated for its potential on neuromorphic hardware.



Literature Survey

Work	Contribution	Relevance to TS-GCN
Kipf & Welling (2017)	Introduced GCN-based graph representation learning	Spatial graph message passing
Neftci et al. (2019)	Surrogate-gradient learning for SNNs	Enables gradient-based SNN training
Tavanaei et al. (2019)	Comprehensive study of deep SNN architectures	SNN design and learning principles
Davies et al. (2018)	Intel Loihi neuromorphic architecture	Hardware motivation for sparse computation
Eshraghian et al. (2023)	Practical SNN training using snnTorch	Implementation framework for TS-GCN



Research Gap

- Existing GNNs mainly focus on spatial relationships and use continuous computation.
- Existing SNNs provide temporal and sparse processing but are less explored for dynamic graph-based fraud detection.
- Few approaches effectively combine graph learning + temporal spiking dynamics for streaming anomalies.
- The potential energy savings and spike sparsity of such models on neuromorphic hardware remain to be evaluated for this application.
- This project addresses the gap through a TS-GCN framework and comparison with a conventional GCN.



Problem Statement

- Dynamic transaction and network graphs contain continuously changing patterns that are difficult to process efficiently.
- Conventional GNNs rely on continuous computation and may be sensitive to short-term fluctuations.
- SNNs offer sparse, event-driven processing but are less explored for graph-based fraud and anomaly detection.
- The project aims to combine GNN spatial learning with SNN temporal dynamics through TS-GCN.
- The proposed model will be compared with a conventional GCN to study improvements in detection performance, computational efficiency, and energy usage.



Hardware & Software Requirements Specification

- Simulation Hardware: Modern laptop/desktop, 8–16 GB RAM, and an NVIDIA CUDA-capable GPU for model training and SNN simulation.
- Simulation Software: Python, PyTorch, PyTorch Geometric, snnTorch, NumPy, Pandas, Scikit-learn, and CUDA.
- Development Tools: Jupyter/Google Colab for experimentation and Git/GitHub for version control.
- Neuromorphic Hardware: For real deployment, platforms such as Intel Loihi 2, SynSense, or BrainScaleS would be required to exploit event-driven computation.
- Event-Based Sensors: Neuromorphic cameras/DVS sensors can be used for event-stream applications, although they are not required for the financial/network transaction use case; specialized hardware is difficult to source, so the project will initially rely on software simulation.



System Design & Methodology

- Data Processing: Convert transaction/network records into a dynamic graph, with entities as nodes and interactions as edges.
- Graph Learning: Use PyTorch Geometric for spatial message passing and feature aggregation across connected nodes.
- Spiking Layer: Pass the aggregated features through LIF neurons to integrate information over time and generate sparse spikes.
- Training: Use surrogate-gradient learning and BPTT to train the non-differentiable spiking network.
- Evaluation: Compare TS-GCN with a conventional GCN using accuracy, F1-score, ROC-AUC, spike sparsity, latency, and estimated energy consumption.



Expected Outcomes

- A working TS-GCN model combining graph convolution with LIF-based spiking dynamics.
- Improved handling of temporal noise and minority-class anomalies compared with a conventional GCN.
- High spike sparsity, reducing unnecessary neural activity during processing.
- Potentially lower computational and energy requirements, particularly on neuromorphic hardware.
- A comparative evaluation of accuracy, F1-score, ROC-AUC, latency, spike sparsity, and estimated energy consumption.



References

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Project Outputs

- Planned: Preparation and submission of a research paper to a peer-reviewed conference/journal on temporal spiking GNNs for energy-efficient dynamic anomaly detection.
- Potential: Development of a patentable TS-GCN framework combining graph learning, temporal spiking dynamics, and energy-efficient anomaly detection, subject to novelty and prior-art evaluation.
- Planned: A reproducible software prototype and simulation framework for TS-GCN experimentation and benchmarking.
- Potential: Further development toward neuromorphic hardware deployment and proof-of-concept demonstration on suitable platforms.
- Future Scope: Exploration of industry collaboration or funded research for real-world deployment and hardware validation.



THANK YOU